

INTERMEDIATE QUANTS 2

Directions for question 1: Type in your answer in the input box provided in the question.

A trader sold an article at 25% profit after giving a 20% discount. If he marked the article at Rs.500, the price at which he bought the article is Rs.

Your Answer: Unattempted

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Solution:

Let the price at which the person bought the article be ₹x

$$\text{Profit made} = \frac{25}{100}x$$

$$\text{Selling price} = x + \frac{25}{100}x = \frac{5}{4}x$$

$$\text{Discount offered} = \frac{20}{100}(\text{MP}) = \frac{20}{100}(500) = 100$$

$$\text{Selling price} = \text{MP} - \text{Discount} = 400$$

$$\frac{5}{4}x = 400$$

$$x = 320$$

Ans : (320)

Directions for questions 2 and 3: Select the correct alternative from the given choices.

If $3x - 4y - 5m = 0$ and $8x - 7y + 2m = 0$, the ratio of x and y is _____.

- ☐ a) $\frac{53}{59}$
- ☐ b) $\frac{91}{99}$
- ☐ c) $\frac{33}{47}$

☒ d) $\frac{43}{46}$ (Your answer - ✓)

Solution:

$$3x - 4y - 5m = 0 \quad \dots (1)$$

$$8x - 7y + 2m = 0 \quad \dots (2)$$

$$2(1) + 5(2) : 46x - 43y = 0$$

$$\frac{x}{y} = \frac{43}{46}$$

Choice (D)

Directions for questions 2 and 3: Select the correct alternative from the given choices.

The sides of a triangle are a , b , c . If 2

$$\sqrt{3}$$

$$a = 2b =$$

$$\sqrt{3}$$

c , the difference of the two smaller angles of the triangle is _____.

- ☐ a) 15°
- ☐ b) 45°
- ☐ c) 30°
- ☐ d) 37.5°

Solution:

$$2\sqrt{3}a = 2b = \sqrt{3}c. \text{ let each of these be } k.$$

$$a = \frac{k}{2\sqrt{3}}, b = \frac{k}{2}, c = \frac{k}{\sqrt{3}} \therefore a : b : c = 1 : \sqrt{3} : 2.$$

$\therefore$ Angles of the triangle are $30^\circ, 60^\circ, 90^\circ$.

Difference of the two smaller angles is 30° .

Choice (C)

Directions for question 4: Type in your answer in the input box provided below the question.

After the annual salary revision exercise in an office, it is found that the average salary of the managers increased by 26.5%, while that of the executives increased by 20%. If the average salary of managers and executives together increased by 22.5% and there are 40 executives in the office, how many managers are there?

Your Answer: Unattempted

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Solution:

The average salary of the managers increased by 26.5%.
The average salary of the executives increased by 20%.
The average salary of both categories of employees increased by 22.5%.
This is closer to the executives average. Therefore, there are more executives.
The ratio of the number of managers to the number of executives = $\frac{22.5 - 20}{26.5 - 22.5}$
$$= \frac{2.5}{4} = \frac{5}{8} = \frac{25}{40}$$

As there are 40 executives, the number of managers is 25. Ans : (25)

Directions for questions 5 and 6: Select the correct alternative from the given choices.

If $x^2 - cx + c^2 - 6c + 12 = 0$ has real roots, then c can take _____ value(s).

- ☐ a) 0
- ☐ b) 1
- ☐ c) 2
- ☐ d) More than 2

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Solution:

Discriminant = $c^2 - 4(c^2 - 6c + 12) = 3c^2 + 24c - 48 = -3(c - 4)^2$
Both roots are real.
 $\therefore$ Discriminant is non-negative.
 $-3(c - 4)^2 \geq 0$
 $(c - 4)^2 \leq 0$ i.e., $(c - 4)^2$ is negative or zero.
But $(c - 4)^2$ cannot be negative.
 $\therefore (c - 4)^2 = 0$.
 $\therefore c - 4 = 0$ i.e. $c = 4$ Choice (B)

Directions for questions 5 and 6: Select the correct alternative from the given choices.

A solid cylinder of radius 5 cm and height 11.25 cm is melted and recast into a hemisphere. Find the radius of the hemisphere.

- ☐ a) 7.5 cm
- ☐ b) 8 cm
- ☐ c) 8.5 cm
- ☐ d) 9 cm

Solution:

Let the radius and the height of the cylinder be r and h respectively and the radius of the hemisphere be R .

$$\pi r^2 h = \frac{2}{3} \pi R^3 \Rightarrow R^3 = \frac{3r^2 h}{2} = \frac{3}{2} (25) \frac{(45)}{4} \text{ cm}^3$$

$$= \frac{(27)(125)}{8} \text{ cm}^3$$

$$\therefore R = \frac{3(5)}{2} \text{ cm} = 7.5 \text{ cm.}$$

Choice (A)

Directions for question 7: Type in your answer in the input box provided below the question.

How many numbers having at most 4 digits can be formed using each of the digits 0, 1, 2, 3 at least once?

Solution:

The single digit numbers which can be formed are 0, 1, 2, 3

The two digit numbers which can be formed are 10, 20, 30, 11, 21, 31, 12, 22, 32, 13, 23, 33.

The three digit numbers which can be formed are of the form abc where a is 1, 2 or 3 and each of b and c ranges from 0 to 3. $\therefore$ Number of three digit numbers which can be formed = $3(4)(4) = 48$

The four digit numbers which can be formed are of the form $abcd$ where a is 1, 2, or 3 and each of b , c , d ranges from 0 to 3. $\therefore$ Number of four digit numbers which can be formed = $3(4)(4)(4) = 192$

Number of numbers which can be formed = $4 + 12 + 48 + 192 = 256$

Ans : (256)

Directions for question 8: Select the correct alternative from the given choices.

$S = \{a, b, c, d, e, f\}$. The number of subsets of S which contain the element a is _____.

- ☐ a) 21
- ☐ b) 40
- ☐ c) 32
- ☐ d) 27

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Solution:

Any subset of S which has a as its element may or may not have each of the remaining five elements of S . $\therefore$ This subset can be formed in 2^5 ways i.e. 32 ways. 32 subsets of S have a as one of their elements. Choice (C)

Directions for question 9: Type in your answer in the input box provided in the question.

A sum of Rs.10000 was lent at 20% p.a. compound interest for 1 year, interest being compounded annually. Had the interest been compounded half yearly, the additional interest realised would be Rs.

Your Answer: Unattempted

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Solution:

$$\text{Interest realised } (I_1) = 10000 \left(\frac{20}{100} \right) = 2000$$

$$\begin{aligned} \text{In the conditional case, the interest realized } (I_2) &= 10000 \left[\left(1 + \frac{20}{2 \times 100} \right)^2 - 1 \right] \\ &= 10000(1.21 - 1) = 2100 \\ \text{Extra interest realized} &= I_2 - I_1 = 100. \end{aligned}$$

Ans : (100)

4. Summary of Standard Exam Shortcuts

1. 2-Period Compound Rate ($r\%$ each period):

$$CI\% = 2r + \frac{r^2}{100}$$

2. 3-Period Compound Rate ($r\%$ each period):

$$CI\% = 3r + \frac{3r^2}{100} + \frac{r^3}{1000}$$

3. Difference between CI and SI for 2 years (Annual $R\%$):

$$CI - SI = P \left(\frac{R}{100} \right)^2$$

4. Difference between CI and SI for 3 years (Annual $R\%$):

$$CI - SI = P \left(\frac{R}{100} \right)^2 \times \left(3 + \frac{R}{100} \right)$$

Directions for questions 10 to 15: Select the correct alternative from the given choices.

The time taken by C to complete a work is half the time taken by A and B together to complete it.

The time taken by B to complete the work is thrice the time taken by A and C together to complete it. The ratio of the rates of work of A and B is _____.

- ☐ a) 1 : 2
- ☐ b) 2 : 1
- ☐ c) 3 : 1
- ☐ d) 1 : 3

Solution:

Let the rates of work of A, B and C be a, b, c respectively.

The rate of C is twice the combined rate of A and B. the rate of B is one – third of the combined rate of A and C.

$$c = 2(a + b) \text{ and } b = \frac{1}{3}(a + c)$$

$$\therefore c = 2(a + b) \text{ and } c = 3b - a$$

$$c = 2(a + b) = 3b - a$$

$$\Rightarrow b = 3a$$

$$\frac{a}{b} = \frac{1}{3}$$

Choice (D)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

The harmonic mean of the roots of a quadratic equation $ax^2 + bx + c = 0$ is

$$\frac{14\sqrt{3} + 30}{5 + 7\sqrt{5}}$$

. Find

$$\frac{b}{c}$$

- ☐ a) $\frac{5+7\sqrt{5}}{-7\sqrt{3}-15}$
- ☐ b) $\frac{-7\sqrt{3}-15}{5+7\sqrt{5}}$
- ☐ c) $\frac{5+7\sqrt{5}}{2(-7\sqrt{3}-15)}$
- ☐ d) $\frac{-7\sqrt{3}-15}{2(5+7\sqrt{5})}$

Solution:

Let the roots be α and β

$$HM(\alpha, \beta) = \frac{14\sqrt{3} + 30}{5 + 7\sqrt{5}}$$

$$\frac{2\alpha\beta}{\alpha + \beta} = \frac{14\sqrt{3} + 30}{5 + 7\sqrt{5}}$$

$$\frac{2\frac{c}{a}}{\frac{-b}{a}} = \frac{14\sqrt{3} + 30}{5 + 7\sqrt{5}}$$

$$\frac{-2c}{b} = \frac{14\sqrt{3} + 30}{5 + 7\sqrt{5}}$$

$$\frac{c}{b} = \frac{-7\sqrt{3} - 15}{5 + 7\sqrt{5}}$$

$$\frac{b}{c} = \frac{5 + 7\sqrt{5}}{-7\sqrt{3} - 15}$$

Choice (A)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

What is the remainder when $5! + 6! + \dots + 20!$ is divided by 9?

- ☐ a) 2
- ☐ b) 0
- ☐ c) 6
- ☐ d) 3

Show Correct Answer

Hide Solution

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Solution:

Every term except the first is divisible by 9. Therefore the remainder is the remainder of the first term.

$$\text{Rem } \frac{5!}{9} = \text{Rem } \frac{120}{9} = 3.$$

Choice (D)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

The radius of the circle circumscribing the triangle whose vertices are (0, 0), (5, 5

$\sqrt{3}$

) (10, 0) is _____.

- ☐ a) $\frac{10}{\sqrt{3}}$
- ☐ b) $\frac{5}{\sqrt{3}}$
- ☐ c) $\frac{2}{2\sqrt{3}}$
- ☐ d) $\frac{15}{4\sqrt{3}}$

Solution:

Any two of the three vertices are 10 units apart.

$\therefore$ The triangle is equilateral and has a side of 10. Radius of the circle circumscribing the triangle = $\frac{10}{\sqrt{3}}$

($\because$ The radius of the circle circumscribing an equilateral triangle = $\frac{a}{\sqrt{3}}$)

Choice (A)

Directions for questions 10 to 15: Select the correct alternative from the given choices.

If $\log_{x-y} x + y = 6$,

$\log_{x^2-y^2}$

$(x + y)^2 = \underline{\hspace{2cm}}$.

- ☐ a) $\frac{12}{7}$
- ☐ b) $\frac{9}{7}$
- ☐ c) $\frac{15}{7}$
- ☐ d) $\frac{6}{7}$

Solution:

$$\log_{x-y} (x + y) = 6,$$

$$2 \log_{x^2-y^2} (x + y) = \frac{2 \log_b x + y}{\log_b x^2 - y^2}$$

where b is any common base

Let $b = x + y$

$$\log_{x-y} (x + y)^2 = \frac{2 \log_b x + y}{\log_b x^2 - y^2}$$

$$= \frac{2(1)}{\log_{x+y} (x + y) + \log_{x+y} (x - y)}$$

$$= \frac{2}{1 + \frac{1}{\log_{x-y} x + y}} = \frac{2}{1 + \frac{1}{6}} = \frac{12}{7}$$

Choice (A)